

Kieran Todd

Mechanical Engineering Student | Power and Energy Systems

Location: Clayton, VIC
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Portfolio: kieran-todd.github.io



About Me:

Third-year Mechanical Engineering student at Monash University, minoring in Power and Energy Systems. Experience across CAD design, electronics integration, and thermodynamic testing and simulation, alongside power system evaluation. Seeking a vacation placement, with a view toward a future career in mechanical and hardware engineering, applying technical and analytical skills to systems where mechanical and electrical components must perform reliably under demanding conditions.

Skills:

Technical Skills: SolidWorks (CSWA Certified), Python, MATLAB, Arduino, CAD Design, Modelling
Professional Skills: Analysis, Problem Solving, Communication, Time Management, Adaptability, Teamwork

Education:

Bachelor of Mechanical Engineering (Honours) | WAM: 90.49 - Monash University *2024 - Expected 2027*

- Terry F. Berreen Award for excellence in Engineering Dynamics (2025)
- Minoring in Power and Energy Systems

Victorian Certificate of Education | ATAR: 99.2 - Bacchus Marsh Grammar *2018-2023*

- Dux of Mathematical Methods, Specialist Mathematics, and Physics

Engineering Projects:

Warman Design & Build Competition | Team-based engineering design and construction challenge

- Designed and manufactured mechanical components in CAD for a team build-and-race competition, working within strict material and cost constraints.
- Applied core mechanical design principles to produce parts that were functional and manufacturable on the workshop's available equipment.
- CAD design was singled out by the workshop lecturer as an example of good practice for the rest of the class.

Power System Simulation Projects | University Group Projects

- Simulated and analysed electrical power systems using PowerWorld and DiGSILENT PowerFactory.
- Investigated voltage control methods and compared their effectiveness at maintaining system stability.
- Modelled the integration of photovoltaic (PV) generation and its impact on network operation.
- Worked in a team to diagnose network issues and develop solutions based on simulation results.

Additional Projects

Ongoing personal and university projects involving mechanical design, CAD, electronics, programming and prototyping. More examples are available in my portfolio.

Work Experience:

Private Tutor | Tutores

2024 - Current

- Provide personalised mathematics and physics tutoring to VCE students, tailoring lessons and resources to individual learning needs and goals.
- Track student progress, give regular feedback, and develop supplementary homework materials.
- Maintain effective communication with students, parents, and staff.
- Received strong recommendations from VCE students' parents.

Customer Service | McDonald's

2020 - 2024

- Mentored new staff to support effective onboarding.
- Managed orders, cash transactions and maintained workplace health and safety standards.